

Topic # 12

User Stories & Estimations in Agile (Scrum)

- REFERENCE Guide: SCRUM & AGILE

- ***User stories are part of an agile approach that helps shift the focus from writing about requirements to talking about them.***
- ***All agile user stories include a written sentence or two and, more importantly, a series of conversations about the desired functionality.***



What is a User Story?

- **User stories** are short, simple descriptions of a feature told from the perspective of the person who desires the new capability, usually a user or customer of the system. They typically follow a simple template:

  As a < type of user >, I want < some goal > so that < some reason >.

- User stories are written in everyday language and describe a specific goal (what) from the perspective of an individual (who) along with the reason (why) he/she wants it.



Role – The user should be an actual human who interacts with the system.
- Be as specific as possible
- The development team is NOT a user

Action – The behavior of the system should be written as an action.
- Usually unique for each User Story
- The "system" is implied and does not get written in the story
- Active voice, not passive voice ("I can be notified")

Benefits – The benefit should be a real-world result that is non-functional or external to the system.
- Many stories may share the same benefit statement.
- The benefit may be for other users or customers, not just for the user in the story



I	Each User Story should represent a distinct and independent set of business values such that, were it released on its own, it would deliver incremental value over the previous state.
N	While the end-goal may be clearly described, the methods by which that goal is achieved should be negotiable – between the Product Owner and the Development Team, the Product Owner
V	The business value of any User Story should be readily recognizable by reading the story, and each story should represent some sort of value to a specific user type
E	We must have enough information that we can properly size a story so that we may properly plan and commit to our work
S	User Stories should be small enough that they are able to be completed within a sprint
T	All members of the team need a clear and precise way to verify whether or not a User Story has been completed

Good Example	Bad Example
• As <a recruiter> I can <post new jobs> so that <applicants can find those jobs via search>	• The software is written in C++
• As <a job seeker> I can <limit who sees my resume> so that <I have privacy>	• The database will have a connection pool

Epic

An **epic** is a large **user story** that cannot be delivered as defined within a single iteration or is large enough that it can be split into smaller **user stories**

Here is an epic agile user story example from a desktop backup product:
As a user, I can backup my entire hard drive.



- The epic above could be split into dozens (or possibly hundreds), including these two:
  - As a power user, I can specify files or folders to backup based on file size, date created and date modified.
  - As a user, I can indicate folders not to backup so that my backup drive isn't filled up with things I don't need saved.

Who writes User Stories?

- Anyone can write user stories.
- It's the product owner's responsibility to make sure a product backlog of agile user stories exists, but that doesn't mean that the product owner is the one who writes them.
- Over the course of a good agile project, you should expect to have user story examples written by each team member.



When to write User Stories

- User stories are written throughout the agile project.
- Usually a story-writing workshop is held near the start of the agile project.
- Everyone on the team participates with the goal of creating a product backlog that fully describes the functionality to be added over the course of the project or a three- to six-month release cycle within it







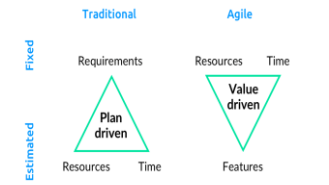
- Write a user story/epic from any of your favorite undergrad projects' module or feature.
- Dos
  - Follow the INVEST criteria
  - Add acceptance criteria to your user story
  - Make it creative & vibrant
- Don'ts
  - No login/logout/user authentication user stories will be accepted :p
  - Avoid copying
  - Every assignment should be different & unique :p

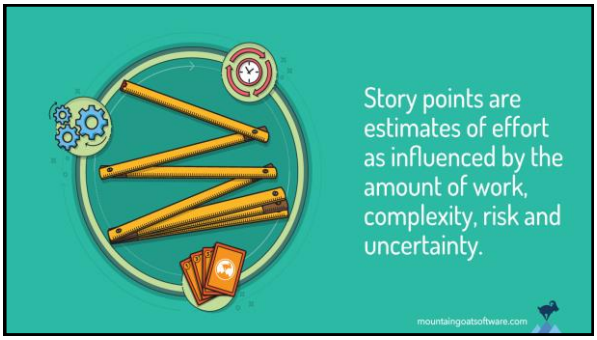
Agile Estimation Techniques



- Agile estimation techniques use a 'top-down' process.
- This encourages teams to propose a gross-level estimation for how long the project should take, or how much effort it will take.
- This is then broken up and applied to different elements of the project.
- Teams drill farther into those elements, uncovering more and more details until the task level – which is looked at through a just-in-time lens.

The paradigm shift





What are Story Points

- Story points are a unit of measure for expressing an estimate of the overall effort that will be required to fully implement a product backlog item or any other piece of work.
- When we estimate with story points, we assign a point value to each item
- For Example:

Points	Probable Duration
2	4 hours
4	8 hours
6	12 hours
8	16 hours

Tee Shirt Sizing

- Often Scrum teams encounter the use of t-shirt sizes (Small, Medium, Large, or so on) in estimating user stories.
- T-shirt sizes are an OK approach to getting started with relative estimating, but they suffer from two severe weaknesses:
  1. They aren't additive.
     - You cannot tell a boss you'll be done in 3 mediums, 4 larges, and 2 petites.
  2. Your view of an XL may not match mine.
     - You may think it's 50% bigger than an L; I may think 25% bigger.

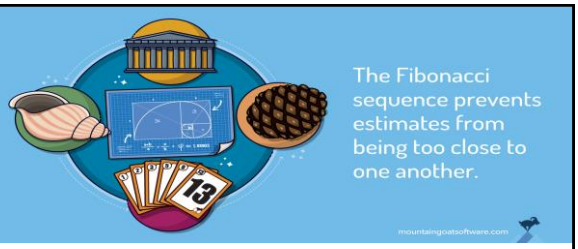




Planning Poker is a consensus-based estimating technique. Agile teams around the world use Planning Poker to estimate their product backlogs. Planning Poker can be used with story points, ideal days, or any other estimating unit.

Planning Poker

- Planning Poker is an agile estimating and planning technique that is consensus based. To start a poker planning session, the product owner or customer reads an agile user story or describes a feature to the estimators.
- Each estimator is holding a deck of Planning Poker cards with values like 0, 1, 2, 3, 5, 8, 13, 20, 40 and 100, which is the sequence we recommend. The values represent the number of story points, ideal days, or other units in which the team estimates.
- The estimators discuss the feature, asking questions of the product owner as needed. When the feature has been fully discussed, each estimator privately selects one card to represent his or her estimate. All cards are then revealed at the same time.
- If all estimators selected the same value, that becomes the estimate. If not, the estimators discuss their estimates. The high and low estimators should especially share their reasons. After further discussion, each estimator reselects an estimate card, and all cards are again revealed at the same time.



- If you've estimated with Planning Poker, you may very well have used cards with either the Fibonacci sequence, or a modified Fibonacci sequence.
- The traditional Fibonacci sequence is 1, 2, 3, 5, 8, 13, 21 and so on, with each number the sum of the preceding numbers.

END OF TOPIC 12

- COMING UP!!!!!!
- Communication & Quality
- Concluding Topics
- Semester Exam
- Project Submission

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